From Retrieval to Orchestration: Building Intent-Driven Context Engineering Systems

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The Future of Data & AI: Agentic AI Conference

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***** Has everyone downloaded Docker and set up the Docker image as well as the installed requirements in a virtual environment as per the ReadMe instructions?******

The problems we will be solving:

Single-turn thinking

Every query = one retrieval call. No context carryover.

Intent blindness

System can't tell what user REALLY wants vs. surface query.

Hallucinated filters

LLM generates filter logic that doesn't match intent.

State collapse

One wrong step cascades. No graceful degradation.

Traditional RAG Architexture

The What

× Hallucinating filters

× Ignoring intent

× No state management

× Single-turn thinking

× Monolithic collapse

× Context blindness

The Why

→ Complex queries

→ Mid-conversation intent switches

→ Multi-step decomposition

→ No observability of failed steps

Lets Start at the beginning

Everything we do
(on the internet)
Starts with Search

Artists, songs, or podcasts

Your top genres

Rock

Pop

IMDb Menu All Search IMDb IMDbPro Watchlist Sign In EN

Up next

4:34

The Latest on 'Black Panther:

amazon.fr Votre adresse de livraison: 75000 Toutes nos catégories

Toutes Meilleures Ventes Amazon Basics Service Client Dernières Nouveautés Ventes Flash Ebooks Kin

Merci Karine, bénévole dans un refuge pour animaux.

Amazon Stars

Idées de cadeaux pour la Saint-Valentin

Découvrir

Vald - Nouvel Album

Découvrir

Les top c

Épicerie

Objets connectés

Meilleures vent

WIKIPEDIA

The Free Encyclopedia

English 6 383 000+ articles

日本語 1 292 000+ 記事

Русский 1 756 000+ статей

Deutsch 2 617 000+ Artikel

Español 1 717 000+ artículos

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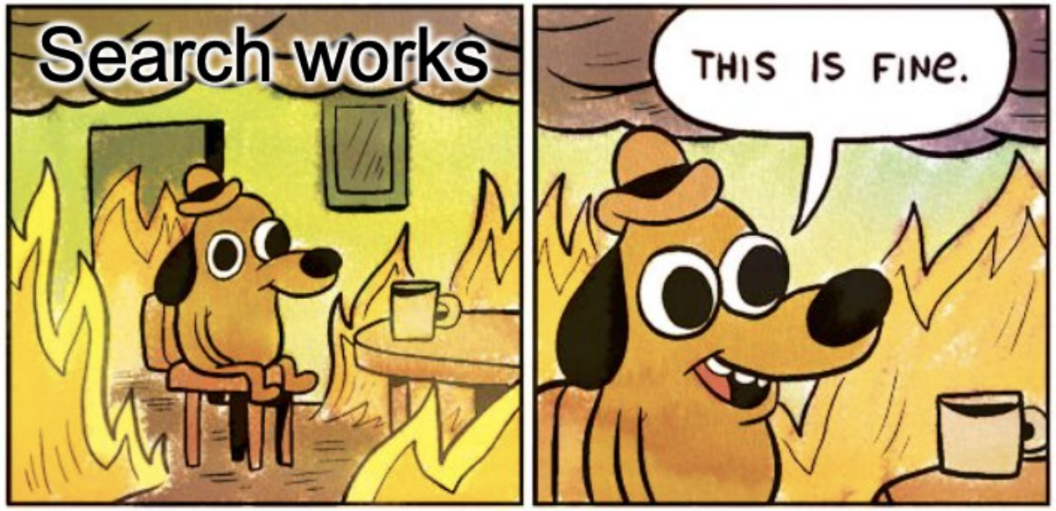
What is the problem?

What is the problem?



The challenges of traditional search

Why do airplanes fly?

Why you should **fly** with ExpensiveAir's **airplanes!**
…fly with ExpensiveAir's modern fleet of **airplanes** to over 4 destinations. **Why do** you hesitate? Book now!

The challenges of traditional search (the keywords match)

Why do airplanes fly?

Why you should **fly** with ExpensiveAir's **airplanes**,
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In summary

Semantic Search to the rescue!

Why do airplanes fly?

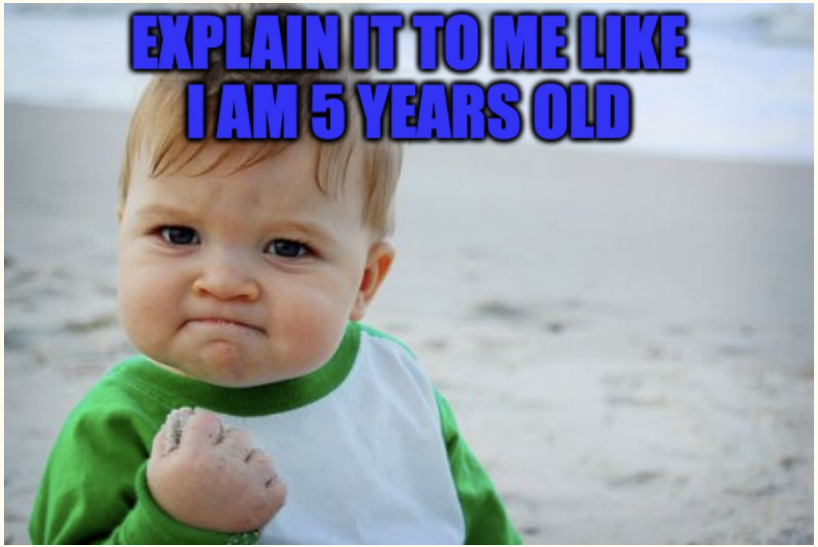
<u>**Dynamics of Flight - NASA**</u>
Airplane wings are shaped **to make air move faster over the top of the wing**. When air moves faster, the pressure of the air decreases.

In summary

How does this work?

How does this work?

"You don't need to understand but it is pretty cool" 😉



How does this work?

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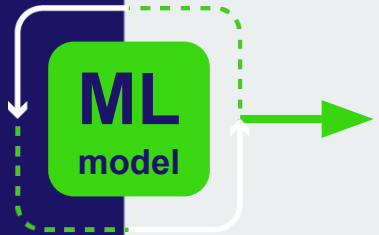
Machine Learning models



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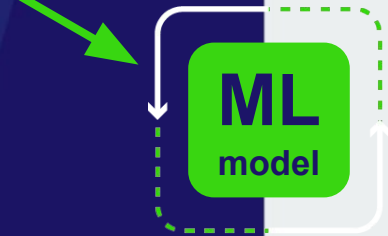
Vector Embeddings

"Fly with ExpensiveAir's modern fleet of airplanes to over 4 destinations. Why do you hesitate? Book now!"



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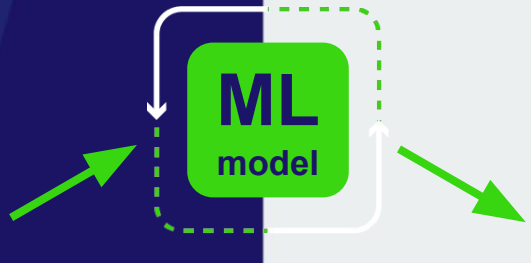

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[0.26, 0.31, ..., 0.12, 0.44]

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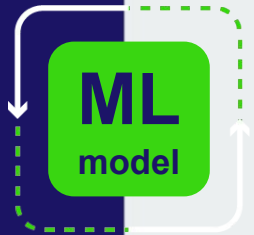

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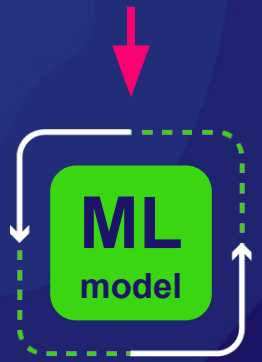
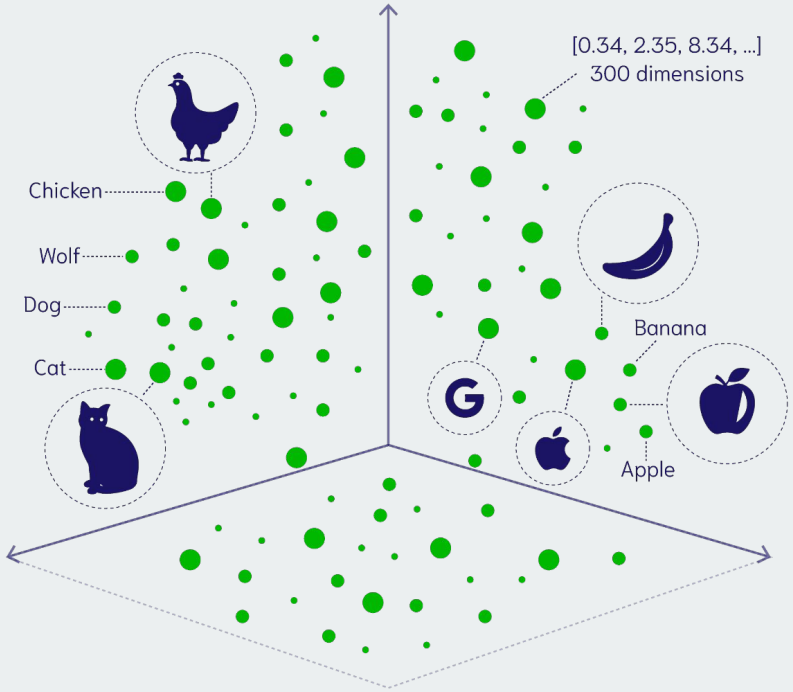
[0.26, 0.31, ..., 0.12, 0.44]

Vector Space

Vector Search

Why do airplanes fly?





Vector Search

Why do airplanes fly?

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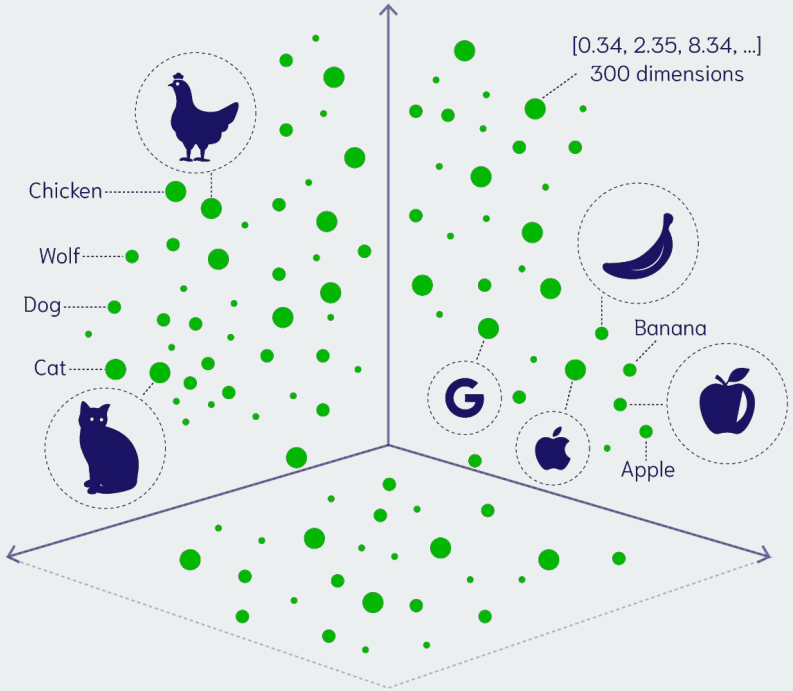

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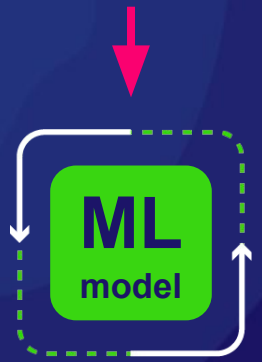

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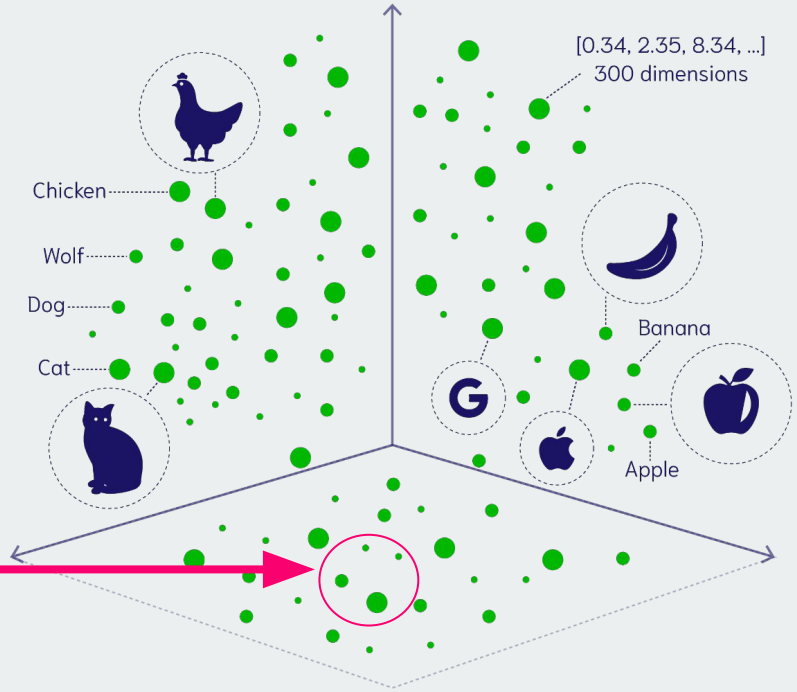

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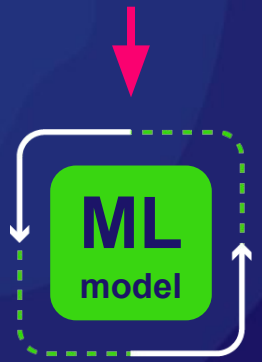

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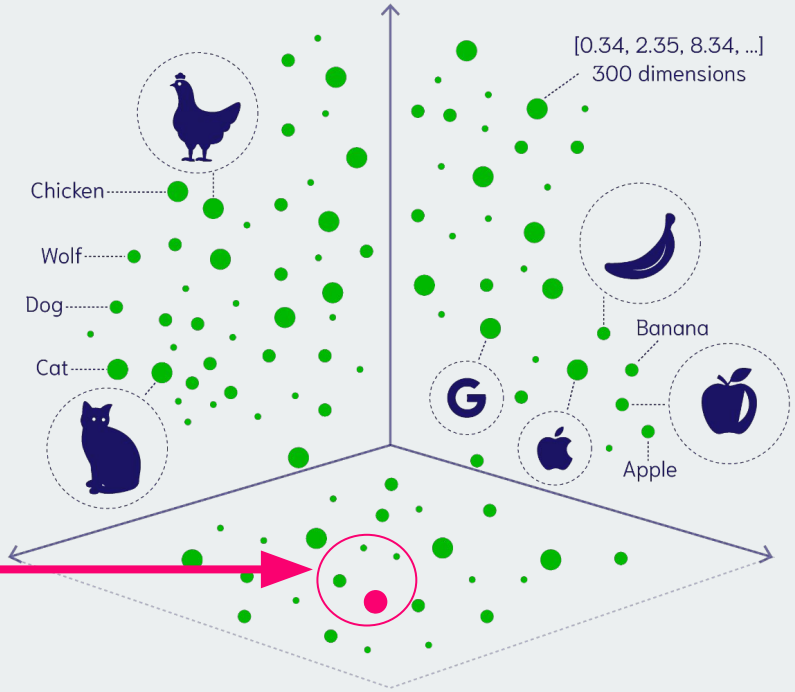
The NASA article



Lesson 1 Vector Database Basics

What is an Agent? (Not Magic)

Old Definition-

LLM with toolcalling

New Definition-

A loop: perceive state → reason about actions → choose action → execute → observe → repeat.

In Our Context-

Agent = structured prompt + tool calls + state management + error handling. **Pydantic AI** gives us the harness.

REACT Framework: Reasoning + Acting

Reasoning -

Think step-by-step BEFORE calling APIs. Decompose intent, evaluate context, plan actions.

Action-

Execute the plan: call vector DB, filters, subagents. Observe results, refine if needed.

Separates 'what should I do' from 'how do I do it'. Enables explainability and debugging.

What is the Recipe for a Good Agent?

1. **An LLM**
2. **Memory**
3. **Well defined tools**

***Bonus* A way to communicate with other agents**

Basic Agentic skills for Vector Databases

1. Well defined, useful skills
   a. The ability to “see” the database properties and metadata
   b. A skill to query the database
      (Hint- these are just functions)
2. Knowledge of how to optimize queries
   a. Natral language simplification
   b. Decomposition

Lesson 2 Agentic Vector Database Queries

How do We Handle Edge Cases?

- A user asks for A, then mid conversation switches to B but also mentions C.
  - Decompose
  - Prioritize
  - Act
- A user has issue A but supplies no information.
  - Determine what was given
  - Seek what is needed

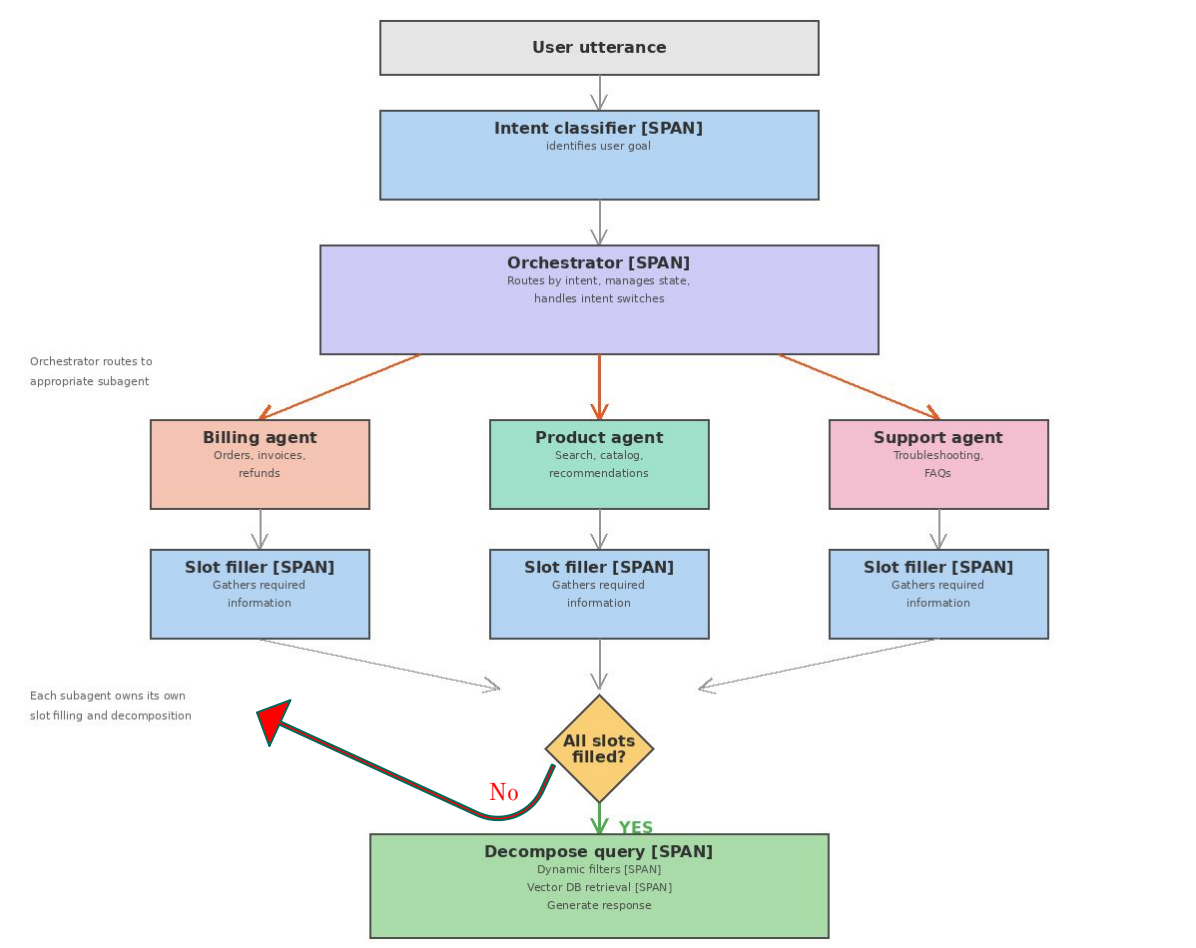
Can an API call to an LLM do any of these things?

Intent-Driven vs. Rank-Based Routing

- **Rank-Based (single agent)-** Retrieve all results → rank by similarity score → pick top-K. Stateless, context-blind.
  -

- **Intent-Based (agentic orchestration)**Classify user intent → route to specialized subagent → that agent decomposes queries → each step informed by intent.

Intent-Based Multi-Agent Architecture



Lesson 3 Agentic Orchestration

BONUS How Do We Know When it Breaks?

- We update our data set
  - Our dynamic filters break
  - Generated answers are incorrect

- We realize our agent isn't pulling customer addresses so we update its prompt
  - Our address slot is not being filled correctly
  - Our generative model is responding with a users address

- We update our Generative Model
  - The model is more confident in its answers
  - We have an uptick in hallucinations

Evaluation Metrics

Retrieval Evaluation:

- **MRR** (Mean Reciprocal Rank)
- **NDCG** (Normalized Discounted Cumulative Gain)
- **Precision** metrics

LLM-Based RAG Evaluation (with Comet Opik):

- **Context Precision**
- **Answer Relevance**
- **Hallucination Detection**
-

Retrieval Evaluation Metrics

MRR (Mean Reciprocal Rank)

- Measures the rank position of the first relevant result (1/position, averaged across queries)
- High MRR but low recall means your retriever finds the right chunks but ranks them too low—add a reranker

NDCG (Normalized Discounted Cumulative Gain)

- Evaluates ranking quality when documents have different relevance levels (compares your ranking to ideal ranking)
- More sophisticated than MRR; requires human-labeled relevance scores but gives nuanced ranking insights

Precision

- Calculates what percentage of retrieved chunks were actually relevant (relevant chunks / total retrieved)
- High precision = clean, focused context; low precision = noisy results confusing your LLM

Generative Evaluation Metrics

Context Precision

- Uses an LLM to judge which retrieved chunks actually contributed to answering the question (relevant chunks / total retrieved)
- More sophisticated than basic precision—catches cases where you retrieved the right document but the chunk is mostly padding

Answer Relevance

- Evaluates whether the final answer actually addresses what the user asked (generates hypothetical questions for the answer, compares to original question)
- Catches off-topic responses where the answer is faithful to context but doesn't answer the user's actual need

Lesson 5 Evaluation Metrics

Thank you!!!!